

# Noise-Robust Discovery of Governing Equations for Mechanical Systems

## Combining DMD and Sparse Regression

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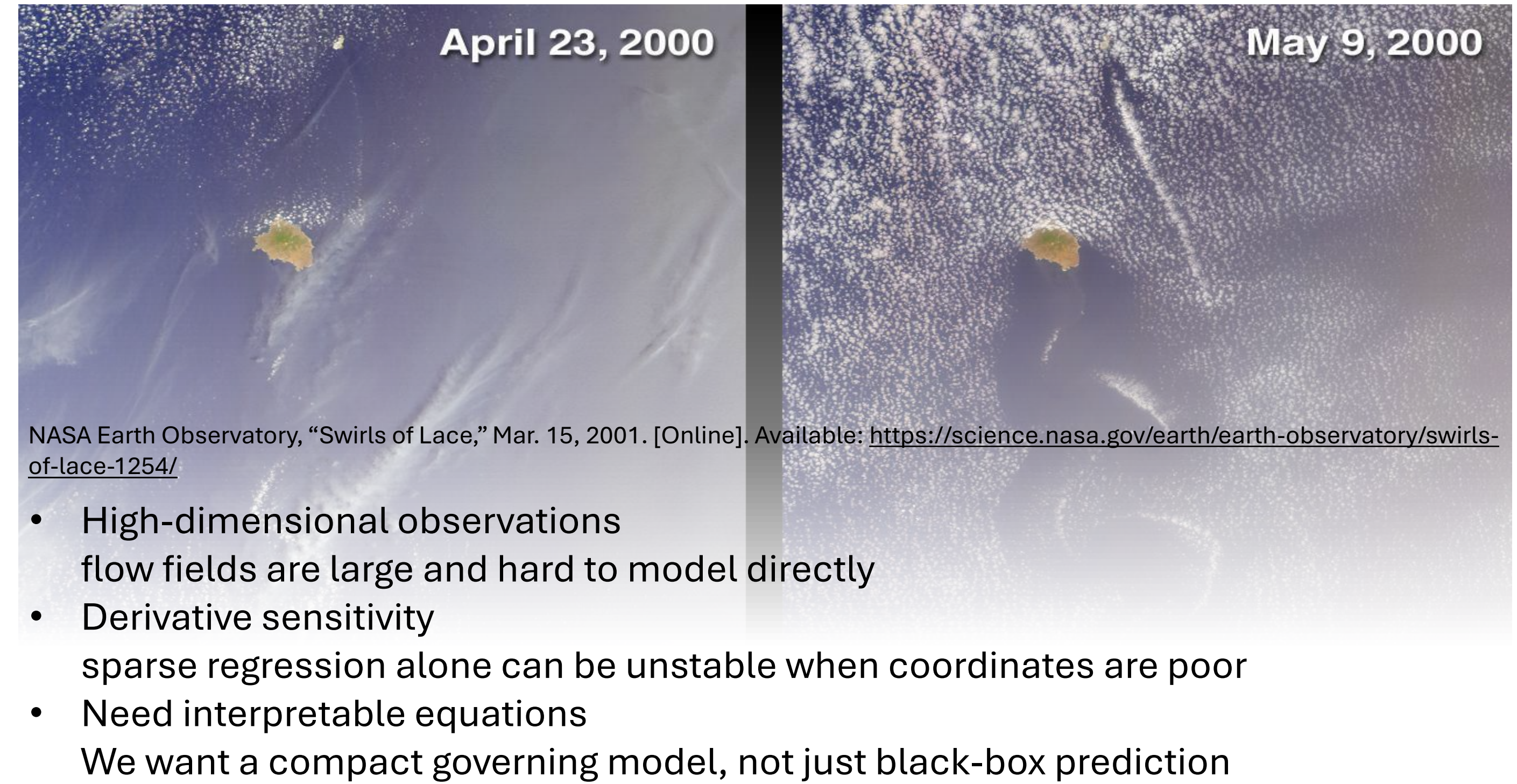
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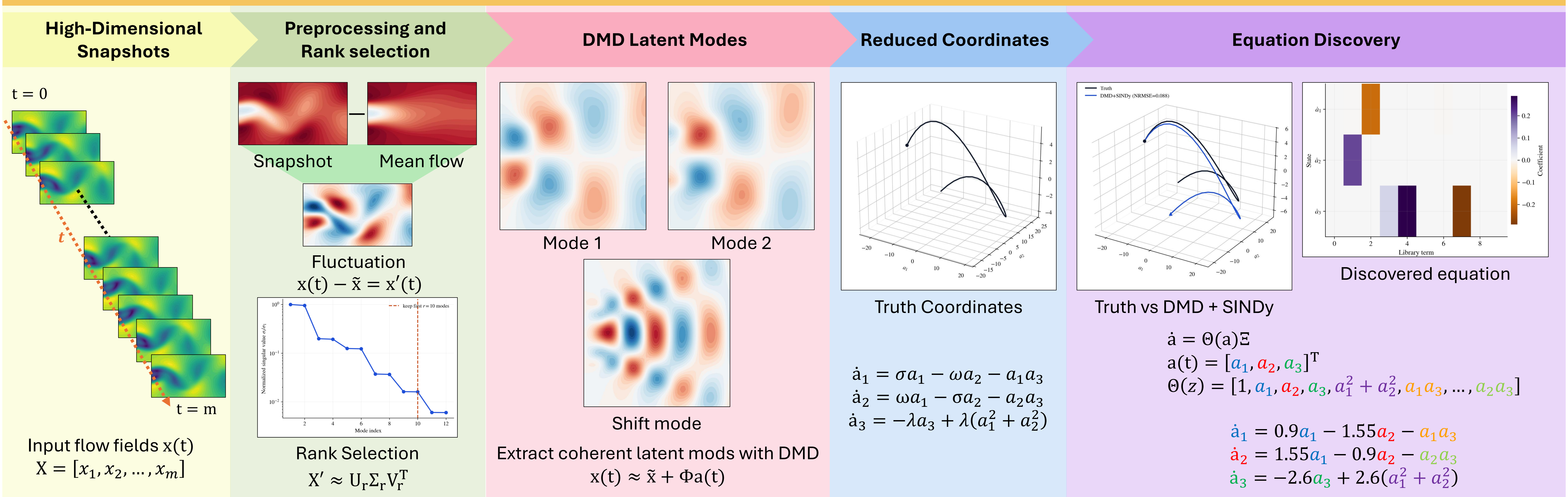
### Abstract

As the Prognostics and Health Management (PHM) field actively tackles the inevitable noise in real-world data, the 'black-box' limitations of widely used deep learning techniques have driven growing attention toward interpretable white-box modeling approaches like the Sparse Identification of Nonlinear Dynamics (SINDy) to ensure essential physical validity and reliability. However, because SINDy relies heavily on numerical differentiation, it suffers from a critical vulnerability. To overcome this limitation, this study proposes a noise-robust pipeline utilizing Dynamic Mode Decomposition (DMD) to discover governing equations. The proposed methodology first applies DMD to high-dimensional, noisy observational data to extract physically meaningful, low-dimensional latent modes, thereby effectively performing dimensionality reduction and noise filtering. Subsequently, with our custom physics-motivated candidate library is applied to these refined latent modes to explicitly formulate the nonlinear dynamic governing equations. The proposed pipeline is validated on nonlinear dynamic benchmarks corrupted with varying levels of noise. Unlike standalone white-box modeling approaches, our model accurately recovers underlying equations even in severely noisy environments. This study presents a practical and interpretable framework for constructing reliable mathematical models directly from complex, noisy mechanical system data to maximize the predictive reliability and explainability of data-driven AI models in engineering applications.

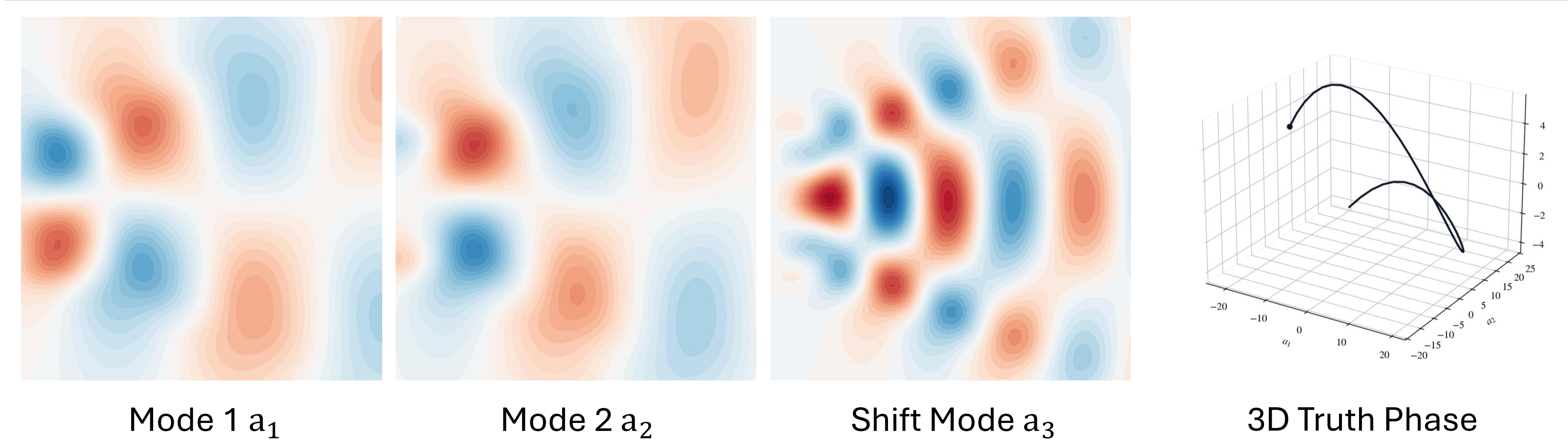
### Why DMD + SINDy?



### Overview



### Reduced Coordinates

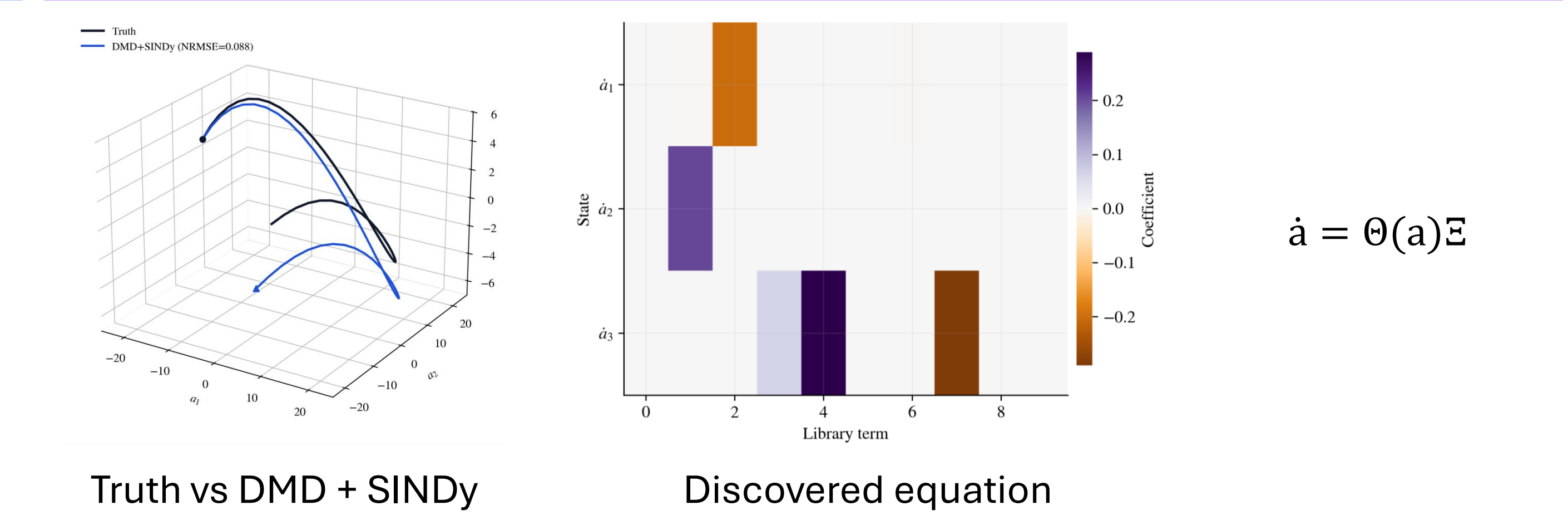


DMD projects the high-dimensional wake snapshots onto a low-dimensional latent space. The reduced coordinates  $a(t) = [a_1, a_2, a_3]^T$  represent the dominant wake dynamics, where  $a_1$  and  $a_2$  describe the oscillatory shedding pair and  $a_3$  captures the shift mode associated with mean-flow deformation.

$$x(t) \approx \tilde{x} + \sum_{i=1}^3 a_i(t) \phi_i$$

$$a(t) = [a_1, a_2, a_3]^T$$

### Equation Discovery



SINDy discovers the governing latent-space dynamics by applying sparse regression to a candidate nonlinear feature library. Only the active terms are retained, yielding a compact and interpretable reduced-order ODE.  
 $\Theta(a)$ : candidate feature library  
 $\Xi$ : Sparse coefficient matrix

### Datasets

#### Kutz Cylinder (vorticity)

- Structured-grid 2D cylinder-wake benchmark
- 151 times snapshots on a  $199 \times 499$  grid
- Contains vorticity fields and reference mean/shift-mode information
- Source: the Brunton-Kutz Data-Driven Science & Engineering companion data. [https://github.com/dynamicsslab/databook\\_matlab](https://github.com/dynamicsslab/databook_matlab)

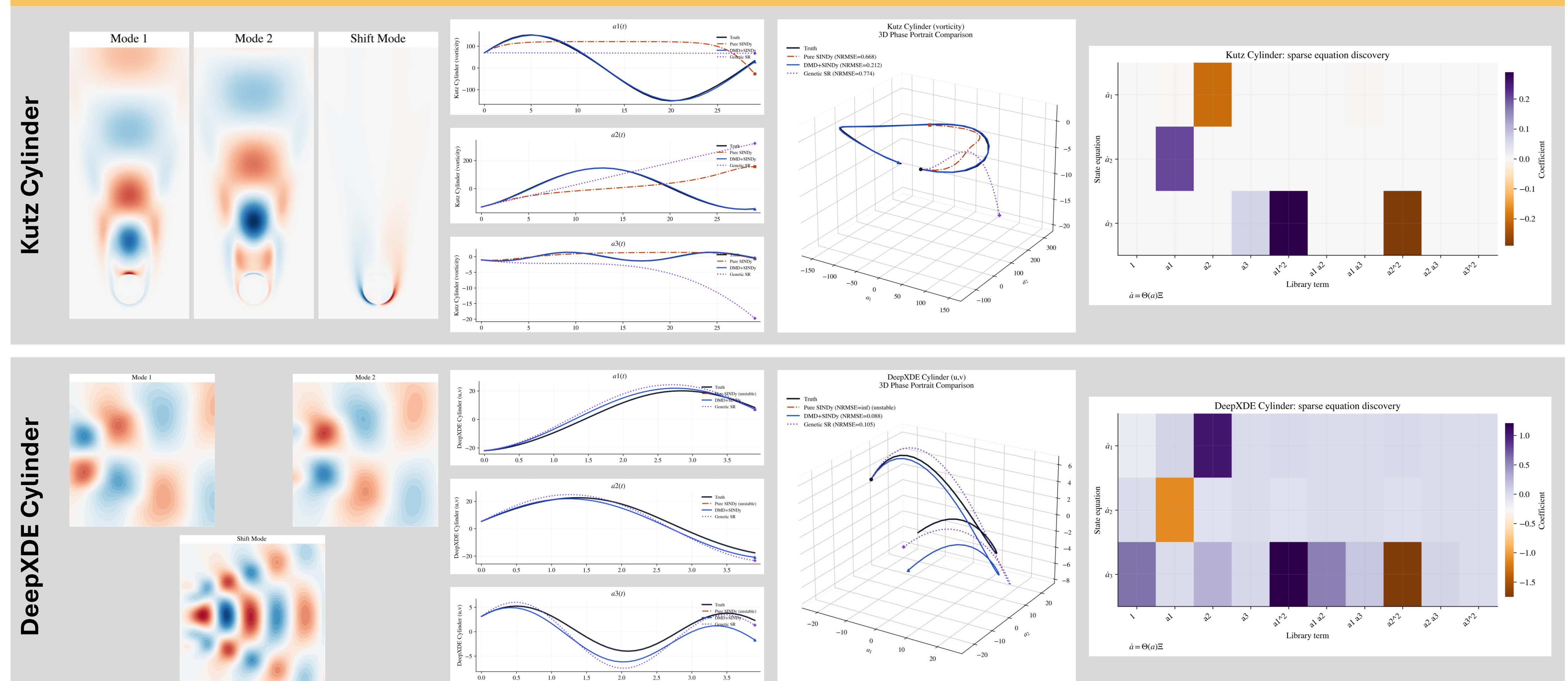
#### DeepXDE Cylinder ( $u, v$ )

- Wake-flow benchmark at  $Re = 100$
- 5000 spatial points and 200 times instants
- Contains velocity fields ( $u, v$ ) on a wake region
- Source: <https://maziarraissi.github.io/PINNs/>

### Conclusion

DMD+SINDy improved reduced-order vortex-shedding reconstruction on both benchmark datasets and produced interpretable latent governing equations. The study further showed that shift-mode recovery is a key requirement for accurate low-dimensional wake modeling.

### Results and Evaluation



### Acknowledgement

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